

# Baselines First

Project: Responsible Plant Classifier

## 1. Your mission

### Your learning target

I can fit a majority label baseline using training labels only.

I can score fixed predictions on the same validation examples.

I can explain what a baseline comparison does and does not establish.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A model must be compared with a simple transparent strategy on the same untouched examples.



LEARNING TARGET A dataset contains 27 healthy and 3 spotted leaves. An always-healthy baseline gets what accuracy and spotted recall? What must a new model improve upon?

Builder name and experiment date:

### Readiness for the connected explanation

Why should frames from the same trip stay in one set when evaluating on new trips?

### This week's end goal

By the end, I can explain baselines first using a traced example and a stated limitation.

## 2. Notice the evidence

PICTURE ENTRY

### Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use baselines first to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

### A closer explanation

A learner reports that a classifier gets 80 percent correct. Is that useful? We need to know the task, the examples and what a simple method achieves on those same examples. A baseline is a reference method. This week our baseline ignores each scene and always predicts the label that is most common in the training set.

Use a new teacher-authored fictional dataset for predicting clear or blocked lanes. Each listed row describes one frame from a different trip on a familiar route; training and validation trips do not overlap. This is a new practice set, not the twelve-frame table from Week 6. All supplied labels have been resolved under the classroom rule. There are no review cases in this exercise, and no real navigation claim is made.

**Set · Labels in row order · Use**

**Set:** Training

**Labels in row order:** clear clear blocked clear blocked clear

**Use:** Choose the constant label

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**Set:** Validation

**Labels in row order:** blocked clear blocked clear

**Use:** Score the fixed predictions

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**Set:** Final test

**Labels in row order:** Not supplied in this exercise

**Use:** Keep for later final evaluation

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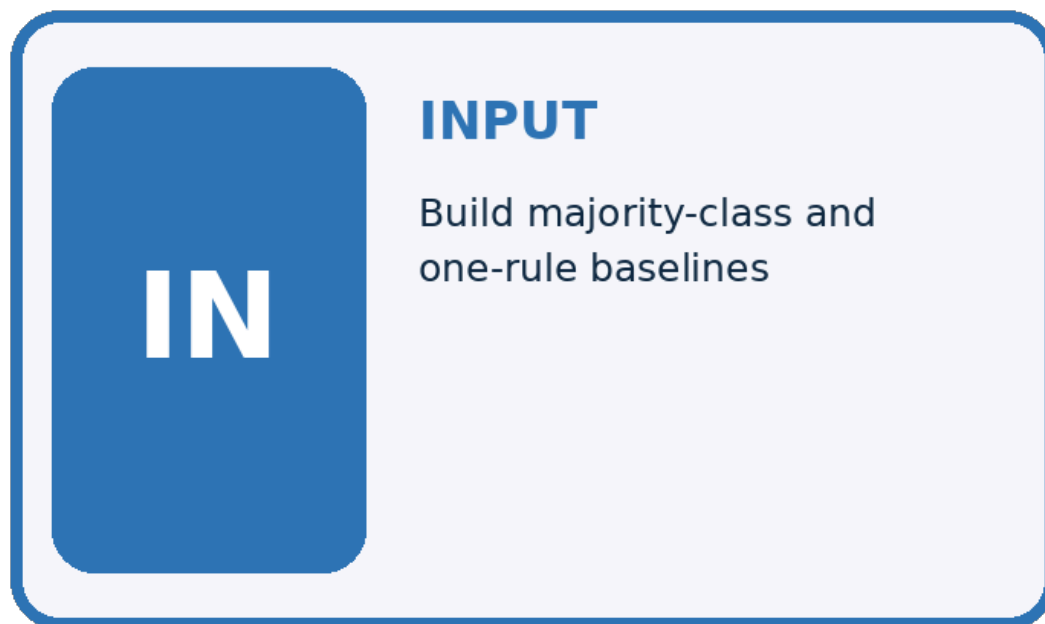
The training counts are four clear and two blocked. Our stated method therefore predicts clear for every new example. A baseline can still learn a value from training data; simple does not mean it uses no data. If the two training counts tie, our rule chooses clear, declared before scoring.

### 3. See how it works

ML MECHANISM

**Input → Model → Evidence**

Trace the information, fitted behavior, and evaluation proof separately.





**ML**

## MODEL

A model must be compared with a simple transparent strategy on the same untouched examples.



**OUT**

## EVIDENCE

then calculate their results.

QUESTION Build majority-class and one-rule baselines, then calculate their results.

MODEL IDEA A model must be compared with a simple transparent strategy on the same untouched examples.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Test set has 12 healthy and 8 spotted leaves. Always-healthy baseline gets  $12 \div 20 = 60\%$  accuracy but 0% spotted recall.

Draw a fresh data → model → prediction → evaluation trace:

## A closer explanation

```
train = ["clear", "clear", "blocked",
        "clear", "blocked", "clear"]
valid = ["blocked", "clear", "blocked", "clear"]
prediction = "clear"

if train.count("blocked") > train.count("clear"):
    prediction = "blocked"

correct = 0

for label in valid:
    if prediction == label:
        correct = correct + 1

print(prediction, correct, len(valid))
```

The output is clear 2 4. The fixed prediction matches two of the four validation labels, so accuracy is 2/4 or 50 percent. The training majority share, 4/6, is not the validation accuracy. The baseline does not inspect the validation labels to decide which constant to predict.

## Keep fitting separate from scoring

### Evidence · Operation · Result

**Evidence:** Training labels only

**Operation:** Count clear and blocked

**Result:** Store prediction clear

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**Evidence:** New validation examples

**Operation:** Repeat stored prediction

**Result:** clear for every row

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**Evidence:** Validation labels

**Operation:** Compare after prediction

**Result:** 2 correct out of 4

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The baseline uses training labels to choose its constant. Validation labels are used to score that fixed constant, not to refit it.

The code assumes nonempty training data with only clear and blocked labels. It counts matches on the supplied validation list; a percentage also requires a nonempty validation set. Do not run it on empty training data and describe the default clear value as learned evidence. Other labels and unresolved review statuses need an explicit policy.

### Connect the picture and the explanation

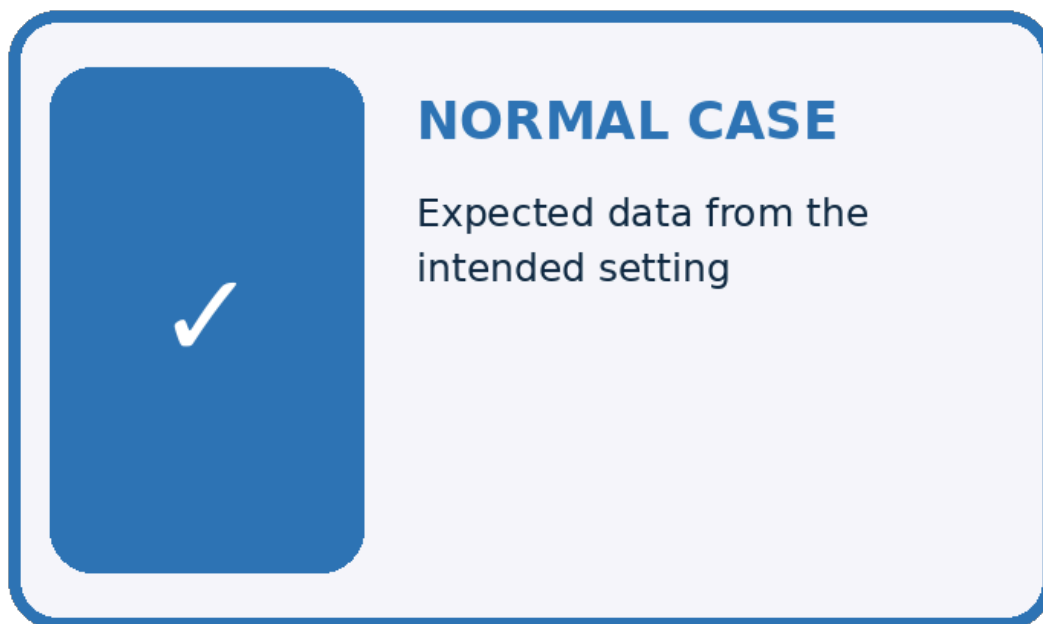
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### EVALUATION DESIGN

#### Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





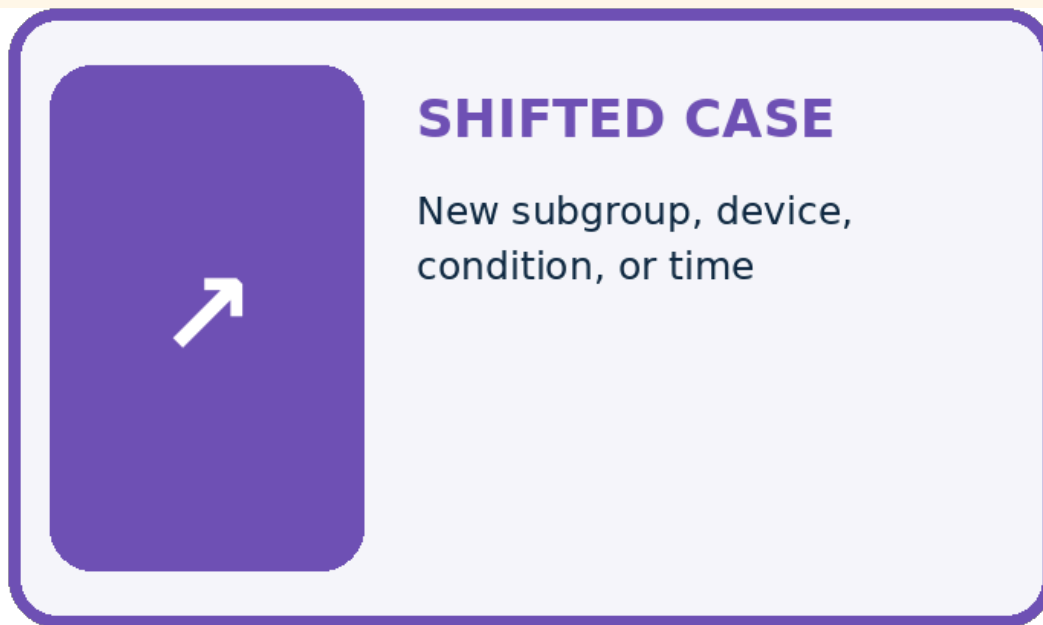
## **BOUNDARY CASE**

Value near a threshold or class boundary



## **MISSING CASE**

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

### Change one thing in the pictured case

Change: Change the third candidate prediction from spotted to healthy.

Prediction: Candidate becomes correct on all 4 validation cases.

Reason: Only the previously wrong third decision changes; baseline stays fixed at 2/4.

## 5. Words that unlock the idea

VOCABULARY

### Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

## **BASELINE**

simple comparison method

2

## **MAJORITY CLASS**

most frequent training label

3

## RULE BASELINE

transparent fixed decision rule

4

## INCREMENTAL VALUE

measured gain beyond baseline

**Term · Precise meaning · My ML example**

**Term:** baseline

**Precise meaning:** simple comparison method

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** majority class

**Precise meaning:** most frequent training label

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** rule baseline

**Precise meaning:** transparent fixed decision rule

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** incremental value

**Precise meaning:** measured gain beyond baseline

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

MISCONCEPTION TO REPAIR Any trained model is more intelligent and therefore better than a simple rule.

Repair the claim using a definition, calculation, or test record:

## Words used in the closer explanation

### Word · Meaning

**Word:** Baseline

**Meaning:** A simple reference method used to judge a more complex approach

\_\_\_\_\_

**Word:** Majority label

**Meaning:** The label with more examples than the other label in this binary task

\_\_\_\_\_

**Word:** Accuracy

**Meaning:** Correct predictions divided by the number of scored examples

\_\_\_\_\_

**Word:** Percentage point

**Meaning:** A unit for the difference between two percentages

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## 6. Follow a worked example

### WORKED EVIDENCE

#### Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

#### Four-step worked evidence for Week 10: Baselines First

##### Step · Evidence and reasoning

##### Step: 1

**Evidence and reasoning:** Test set has 12 healthy and 8 spotted leaves.

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##### Step: 2

**Evidence and reasoning:** Always-healthy baseline gets  $12 \div 20 = 60\%$  accuracy but 0% spotted recall.

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##### Step: 3

**Evidence and reasoning:** On the same 20 leaves, the visible-spot rule correctly labels 5 spotted leaves, misses 3 spotted leaves, correctly labels 10 healthy leaves and falsely marks 2 healthy leaves as spotted. Accuracy= $(5+10)/20=75\%$ ; spotted recall= $5/8=62.5\%$ .

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##### Step: 4

**Evidence and reasoning:** A learned model must beat relevant baseline metrics—not only one convenient total.

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Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

STEP 1 Test set has 12 healthy and 8 spotted leaves.

STEP 2 Always-healthy baseline gets  $12 \div 20 = 60\%$  accuracy but 0% spotted recall.

STEP 3 On the same 20 leaves, the visible-spot rule correctly labels 5 spotted leaves, misses 3 spotted leaves, correctly labels 10 healthy leaves and falsely marks 2 healthy leaves as spotted. Accuracy= $(5+10)/20=75\%$ ; spotted recall= $5/8=62.5\%$ .

STEP 4 A learned model must beat relevant baseline metrics—not only one convenient total.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a

score.

Recalculate, retrace, or restate the evidence independently:

## A closer explanation

Suppose a candidate method supplies blocked, clear, clear, clear for the four validation rows in the table. These are supplied predictions for a comparison exercise; we have not fitted that candidate. It gets three of four correct, or 75 percent. The baseline gets two of four, or 50 percent. The candidate is ahead by one correct example and 25 percentage points on this set.

Use identical row IDs, reference labels, exclusions and metric for both methods. Match predictions to their rows even if files are reordered. A score on easier examples is not a fair comparison. Neither this four-row result nor beating one baseline proves that the candidate will perform well on future trips.

**The candidate misses the third validation row. Which rows does the baseline miss? Explain how the candidate changes the number correct. Why is the improvement 25 percentage points rather than a claim of 25 percent relative improvement?**

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Record training counts, the fixed prediction and tie rule, validation IDs and labels, prediction lists and correct/total counts. Keep the final test separate from development decisions. Next week we will search candidate rules using training evidence, then compare choices carefully.

## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Baselines First

Make a simple baseline earn its place before claiming a better classifier.

## The facts we will use

Training labels: healthy, healthy, spotted. Majority baseline predicts healthy. Validation references: healthy, spotted, healthy, spotted.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Baselines First

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Fit baseline

Training majority is healthy: 2 of 3 labels.



#### 2. Score baseline

Always healthy matches 2/4 validation cases: 50%.



#### 3. Compare candidate

Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

### CHANGE ONE THING

Change the third candidate prediction from spotted to healthy.

Explain the changed result using the matching lesson.

### Read the picture one step at a time

#### Step · What happens and why

**Step:** 1. Fit baseline

**What happens and why:** Training majority is healthy: 2 of 3 labels.

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**Step:** 2. Score baseline

**What happens and why:** Always healthy matches 2/4 validation cases: 50%.

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### Step: 3. Compare candidate

**What happens and why:** Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Read the picture in order. Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

These supplied predictions illustrate comparison. No actual model was trained, and four cases are limited evidence.

## 7. Find and repair a mistake

ERROR DETECTIVE

### Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



#### COMPARE • DIAGNOSE • REPAIR

Tempting claim: Any trained model is more intelligent and therefore better than a simple rule. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** passes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** defined behavior

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** missing case

**Expected:** safe fallback

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** shifted case

**Expected:** measured limit

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**CLAIM TO CHALLENGE** Any trained model is more intelligent and therefore better than a simple rule.

Diagnose one cause and name the cleanest next experiment:

## A closer explanation

Imagine ten validation cases with nine clear and one blocked, while the training-derived constant is clear. The baseline gets 9/10 or 90 percent accuracy but misses the only blocked case. Overall accuracy counts all rows equally; it does not describe every kind of error or its consequences. Later weeks will examine those errors in detail.

A majority baseline is a useful first comparison, not a guarantee of the best simple method. It can be weak when validation label proportions differ from training. Keep the declared baseline, report what happened and consider other justified reference methods during development. Do not tune a final test result into a better-looking claim.

**A candidate and the clear baseline both get 90 percent on ten cases. Does that prove they make the same mistakes or are suitable for use? State what additional evidence you would**

inspect.

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## 8. Practise together

ML STUDIO

### Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



#### **FREEZE THE EXPERIMENT**

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

### Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

### A closer explanation

A learner sees that blocked is common in a new validation set and changes the baseline to always blocked. That is not the original training-majority baseline. Its parameter was selected from evaluation labels. Restore the training-derived constant for that comparison and document any separate method honestly.

Another learner compares a candidate on ten easy rows with a baseline on four difficult rows. The difference could come from the examples. Score both on the same intended set with the same treatment of missing predictions and unresolved labels; report exclusions rather than silently dropping mistakes.

## Practise with less help: Calculate the validation score

Use workbook W2 for the connected example “A baseline before a model”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for A baseline before a model. Proposed explanation: “A simple baseline does not need training data”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Can the candidate’s training accuracy replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

EVIDENCE LAB

**Predict → Run → Calculate → Interpret**

Keep the test conditions fixed, record every result, and change only one planned factor.

**LAB**

**PREDICT → RUN → CALCULATE → INTERPRET**

Baseline Challenge: calculate two baselines on one fixed set, compare class-specific errors, and choose the honest benchmark.

1. Write the hypothesis and expected evidence before running.
2. Baseline Challenge: calculate two baselines on one fixed set, compare class-specific errors, and choose the honest benchmark.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

**Case · Prediction · Actual · Calculation / evidence · Meaning / next test**

**Case:** Normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Shifted

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

**From Knowing the Word to Owning the Idea**

Move upward only when you can produce evidence without copying the example.



## ML BUILDER MASTERY

1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: baseline, majority class, rule baseline, incremental value.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Any trained model is more intelligent and therefore better than a simple rule.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

## 11. Connect your project

CUMULATIVE PROJECT

### Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



## RESPONSIBLE PLANT CLASSIFIER

Record majority and transparent-rule baselines for the Plant Classifier using the frozen test plan.

**THIS WEEK'S CHECKPOINT** Record majority and transparent-rule baselines for the Plant Classifier using the frozen test plan.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

**RESPONSIBLE RELEASE** Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

### Evidence for your project

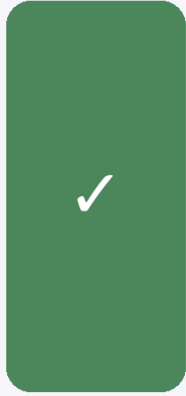
For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 12. Show what you know

INDEPENDENT PROOF

**Explain • Calculate • Transfer • Defend**

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



### INDEPENDENT ML BUILDER PROOF

Build majority-class and one-rule baselines, then calculate their results.

REQUIRED PROOF Build majority-class and one-rule baselines, then calculate their results.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Decision Trees.

## 13. Study a complete case

### Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

A test set has eighteen healthy and two spotted leaves. Baseline always predicts healthy.

#### The question

Calculate and critique accuracy.

#### Worked explanation

18/20=90%, but spotted recall is 0/2=0%. Compare with task-relevant metrics and the trained model.

## Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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## Trace the supplied case visually

### A majority baseline can miss every positive

**Evidence or stage · Value or result**

**Evidence or stage:** Overall correct

**Value or result:** 18/20

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**Evidence or stage:** Spotted found

**Value or result:** 0/2

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A test set has eighteen healthy and two spotted leaves. Baseline always predicts healthy.

Calculate and critique accuracy.

18/20=90%, but spotted recall is 0/2=0%. Compare with task-relevant metrics and the trained model.

## 14. Try a new case independently

### Independent case: Fit and compare independently

New training labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same baseline rule and list its four predictions. A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and percentage points. What claim remains unsupported?

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A dataset has 18 grass and 2 fern examples. A baseline always predicts grass. Compute accuracy and name a serious limitation.

My answer, with a trace, calculation or supporting evidence

#### Write or draw your reasoning and evidence.

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### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

#### Write or draw your reasoning and evidence.

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## Connect to your project

Record majority and transparent-rule baselines for the Plant Classifier using the frozen test plan.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

## Transfer practice from the original programme

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy baseline gets what accuracy and spotted recall? What must a new model improve upon?

**Explain the mechanism, evidence and one limit.**

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## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a

model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.